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Separation Logic for Automated Verification

A Partial Overview

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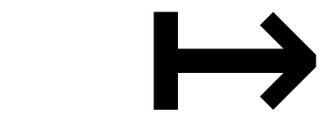

Introduction to Separation Logic

Hoare Reasoning

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Hoare Triple: {P} prog {Q}

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ASSIGN $\{Q[e/x]\}\ x := e\ \{Q\}$

For **local variables** x and y:

$$\frac{\{2 = 2 \wedge y = 3\}\ x := 2\ \{x = 2 \wedge y = 3\}\ \text{(by ASSIGN)}}{\{x = 3 \wedge y = 3\}\ x := 2\ \{x = 2 \wedge y = 3\}\ \text{(by CONSEQ)}}$$

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For **local variables** x and y:

$$\frac{\{2 = 2 \wedge y = 3\}\ x := 2\ \{x = 2 \wedge y = 3\}\ \ (\text{by ASSIGN})}{\{x = 3 \wedge y = 3\}\ x := 2\ \{x = 2 \wedge y = 3\}\ \ (\text{by CONSEQ})}$$

(ASSIGN is) valid (only) because x and y cannot alias.

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The answer is **Separation Logic**

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But before lets see how can we come up with a notion of **memory (aka heap)**,
the memory that pointers are pointing to

Heap Reasoning

What if x and y are pointers into the heap?

*x := 2

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```
void foo(int *x){
    int *y := x;
    *x := 3;
    *x := 2;
}
```

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```

It does't hold when x and y **alias** each other.

Global Reasoning

We need a way to reason **locally** about statements like

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…	…

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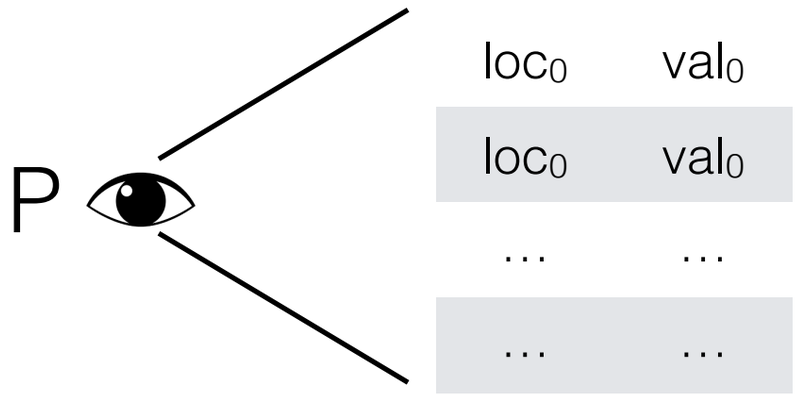

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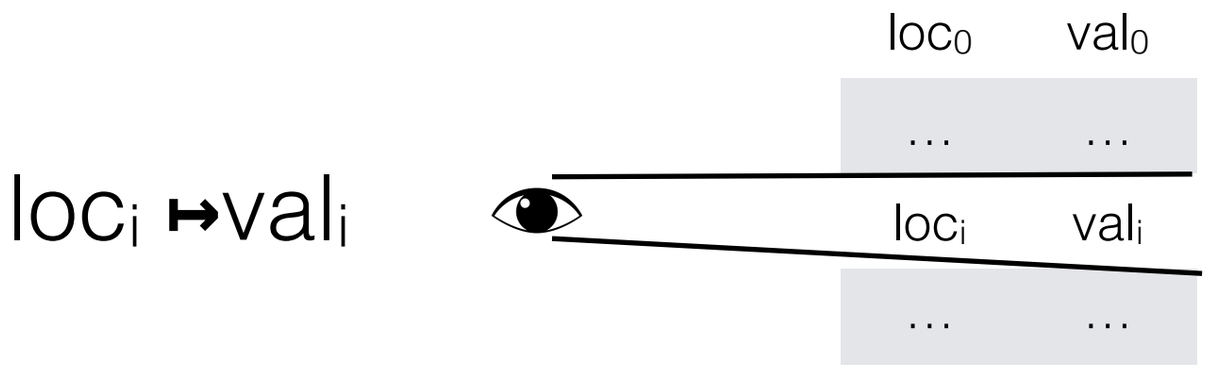

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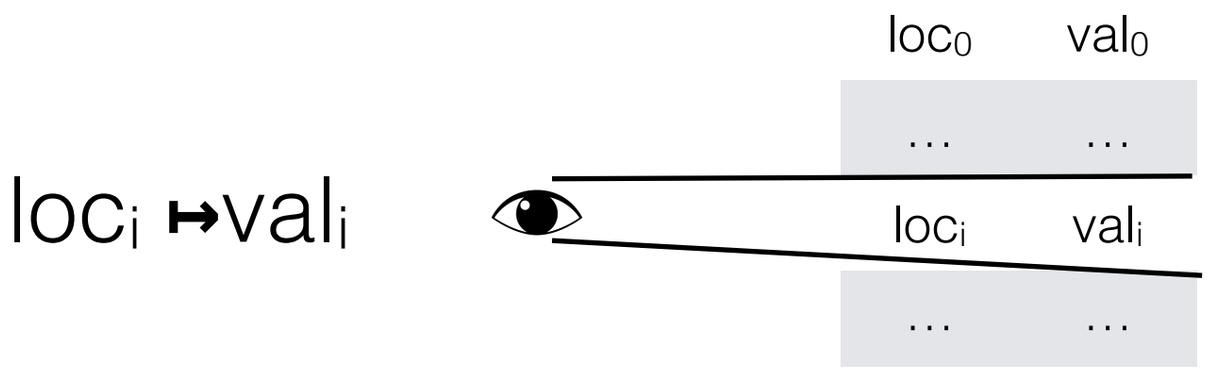

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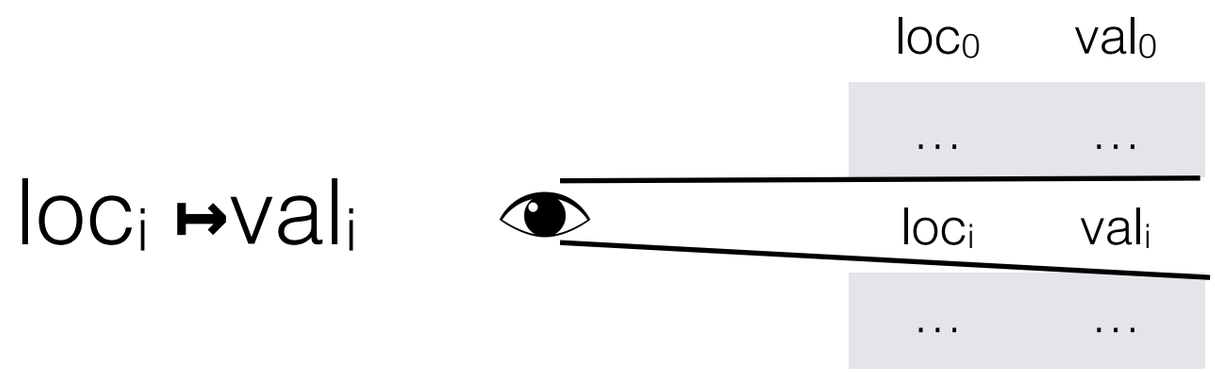

This part is called the predicate's **domain** (aka footprint), written dom(P)

$$dom(p \mapsto e) = \{p\}$$

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A valid triple:

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x	2
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Satisfies $x \mapsto 2$

How do we talk about **other** parts of the heap?

Compositional Reasoning

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Separating Conjunction: $P \star Q$

"P holds for its part of the heap and Q holds for its part and, moreover, **those two parts are separate**."

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Is this triple valid?

$$\{x \mapsto 3 \star y \mapsto 3\}\ {*x} := 2\ \{x \mapsto 2 \star y \mapsto 3\}$$

Yes.

The Frame Rule

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$$\text{FRAME}\ \frac{\{P\}\ prog\ \{Q\} \qquad modv(prog) \cap fv(R) = \{\}}{\{P \star R\}\ prog\ \{Q \star R\}}$$

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modv(prog): the set of local variables modified by prog

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modv(prog): the set of local variables modified by prog

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$modv(*x := 2) = \{\}$ $\qquad$ $fv(y \mapsto 3) = \{y\}$

The Side Condition

$$\text{FRAME}\ \frac{\{P\}\ prog\ \{Q\} \qquad \mathrm{modv}(prog) \cap \mathrm{fv}(R) = \{\}}{\{P \star R\}\ prog\ \{Q \star R\}}$$

$$\mathrm{modv}(*x := 2;\ y := x) = \{y\}$$
$$\mathrm{fv}(y \mapsto 3) = \{y\}$$

Side condition doesn't hold.

The Side Condition

$$\text{FRAME}\ \frac{\{P\}\ prog\ \{Q\} \qquad \mathrm{modv}(prog) \cap \mathrm{fv}(R) = \{\}}{\{P \star R\}\ prog\ \{Q \star R\}}$$

Suppose we had:

$\{x \mapsto 3\}$ *x := 2; y := x $\{x \mapsto 2\}$

$\mathrm{modv}(*x := 2; y := x) = \{y\}$

$\mathrm{fv}(y \mapsto 3) = \{y\}$

Side condition doesn't hold.

The Side Condition

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Suppose we had:

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Clearly, it is **not** the case that:

$$\{x \mapsto 3 \star y \mapsto 3\}\ {*x} := 2;\ y := x\ \{x \mapsto 2 \star y \mapsto 3\}$$

$\mathrm{modv}(*x := 2;\ y := x) = \{y\}$

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Side condition doesn't hold.

Validity of Triples

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Meaning of {P} prog {Q} **in Separation Logic**

Validity of Triples

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Meaning of {P} prog {Q} **in Separation Logic**

Partial Correctness: If P holds initially then whenever prog terminates successfully, then Q holds and, moreover, **prog never fails.**

Memory Safety

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Example: {x = `NULL`} y := *x {true}

Does this triple hold?

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In a language semantics in which all memory-unsafe accesses cause failure,
separation logic guarantees memory safety.

e.g. accessing an uninitialised pointer.

```
int *p;
*p = 2;
```

Expressions, Commands

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Expressions: $e ::= x \mid n \mid \ldots$

May mention (local) variables but not the heap.

Are evaluated against a store s :: Var → Int

$[\![e]\!]_s$ :: Int

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Commands: $c ::= c_1 \,;\, c_2 \mid$ **if** e **then** c_t **else** c_f
$\mid$ **while** e **do** $c \mid x := e$
$\mid x := {}^*e \mid {}^*e_1 := e_2$

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Note: heap- and store-update are syntactically distinct.

Heap Read Rule (Simplified)

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$$\frac{x \notin \text{vars}(e)}{\{e \mapsto v\}\ x := {}^*e\ \{x = v \wedge e \mapsto v\}}$$

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$$x := {*(y+1)}$$

Heap Read Rule (Simplified)

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$s(y) = 19$

h

…	…
4	2
…	…
20	3
…	…

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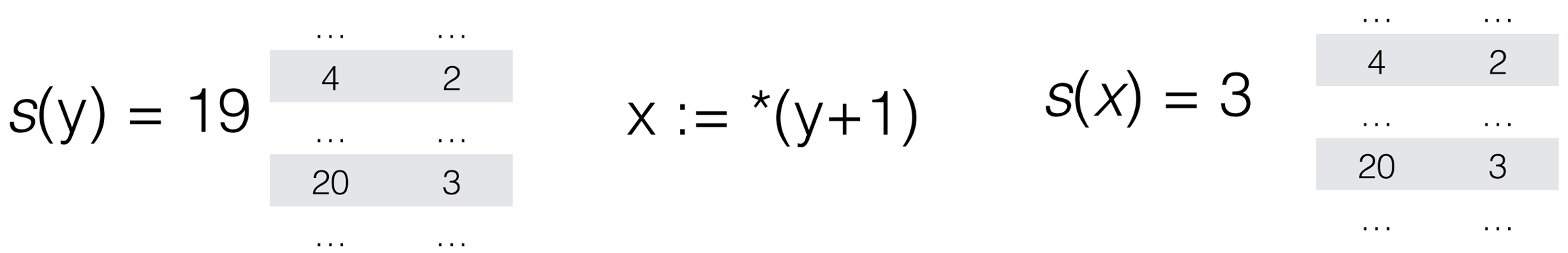

y+1 ↦ 3 holds

NOTE: We need a more general rule for when x ∈ vars(e) but that is beyond the scope of today's lecture

Assignment Rules

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HEAPW $\quad \{e_1 \mapsto \_\}\ {*e_1} := e_2\ \{e_1 \mapsto e_2\}$

ASSIGN $\quad \{x \doteq n\}\ x := e\ \{x \doteq e[n/x]\}$

ASSIGN' $\quad \dfrac{x \notin \mathrm{vars}(e)}{\{\mathbf{emp}\}\ x := e\ \{x \doteq e\}}$

Frame Rule

$$\text{modv}(x := e) = \{x\}$$
$$\text{modv}(x := *e) = \{x\}$$
$$\text{modv}(*e_1 := e_2) = \{\}$$
$$\text{modv}(\textbf{while } e \textbf{ do } c) = \text{modv}(c)$$
$$\text{modv}(\textbf{if } e \textbf{ then } c_t \textbf{ else } c_f) = \text{modv}(c_t) \cup \text{modv}(c_f)$$
$$\text{modv}(c_1 ; c_2) = \text{modv}(c_1) \cup \text{modv}(c_2)$$

$$\text{FRAME}\ \frac{\{P\}\ c\ \{Q\} \qquad \text{modv}(c) \cap \text{fv}(R) = \{\}}{\{P \star R\}\ c\ \{Q \star R\}}$$

Example Proof

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$$\{x \mapsto v_x \star y \mapsto v_y\}$$
$$t := *x;$$
$$u := *y;$$
$$*x := u;$$
$$*y := t;$$
$$\{x \mapsto v_y \star y \mapsto v_x\}$$

Example Proof

$$\{x \mapsto v_x \star y \mapsto v_y\}$$
$$t := *x;$$
$$u := *y;$$
$$*x := u;$$
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$$\{x \mapsto v_y \star y \mapsto v_x\}$$

Apply the Hoare sequencing rule

Example Proof

$$\{x \mapsto v_x \star y \mapsto v_y\}$$
$$t := *x;$$
$$\{(x \mapsto v_x \wedge t = v_x) \star y \mapsto v_y\}$$
$$\{(x \mapsto v_x \wedge t = v_x) \star y \mapsto v_y\}$$
$$u := *y;$$
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Example Proof Outline

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$$*y := t;$$

Example Proof Outline

$$\{x \mapsto v_x \star y \mapsto v_y\}$$
$$t := *x;$$
$$\{(x \mapsto v_x \wedge t = v_x) \star y \mapsto v_y\}$$
$$u := *y;$$
$$\{(x \mapsto v_x \wedge t = v_x) \star (y \mapsto v_y \wedge u = v_y)\}$$
$$*x := u;$$
$$\{(x \mapsto u \wedge t = v_x) \star (y \mapsto v_y \wedge u = v_y)\}$$
$$*y := t;$$
$$\{(x \mapsto u \wedge t = v_x) \star (y \mapsto t \wedge u = v_y)\}$$

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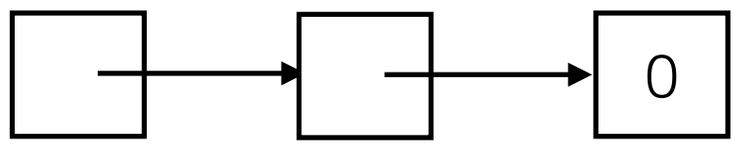
Inductive Predicates

Heap assertions $e_1 \mapsto e_2$ and **emp** talk only about bounded parts of the heap (of size 1 and 0 resp.)

How to talk about unbounded heap data structures?



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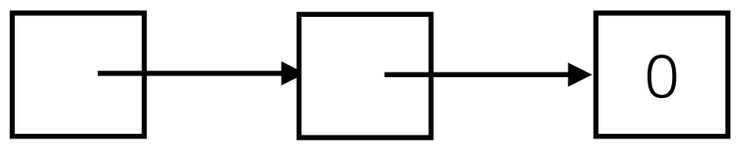

$$\text{list}(e) = (e = 0 \wedge \textbf{emp}) \vee (\exists e'.\ e \mapsto e' \star \text{list}(e'))$$

Example: Tail

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$$\{x \neq 0 \wedge \text{list}(x)\}$$
$$t := *x;$$
$$\{\text{list}(t)\}$$

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Unfold the inductive predicate:

$$\{x \neq 0 \wedge (\exists e'.\ x \mapsto e' \star \text{list}(e'))\}$$
$$t := *x;$$
$$\{\text{list}(t)\}$$

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Unfold the inductive predicate:

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$$t := *x;$$
$$\{\text{list}(t)\}$$

Example: Tail

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$$\{x \neq 0 \wedge x \mapsto e' \star \mathrm{list}(e')\}$$

t := *x;

{list(t)}

Example: Tail

$$\{x \neq 0 \wedge x \mapsto e' \star \text{list}(e')\}$$
$$t := *x;$$
$$\{\text{list}(t)\}$$

$$\{x \neq 0 \wedge x \mapsto e'\}$$
$$t := *x;$$
$$\{x \neq 0 \wedge x \mapsto e' \wedge t = e'\}$$

Example: Tail

$$\{x \neq 0 \wedge x \mapsto e' \star \mathrm{list}(e')\}$$
$$t := *x;$$
$$\{\mathrm{list}(t)\}$$

$$\{x \neq 0 \wedge x \mapsto e'\}$$
$$t := *x;$$
$$\{x \neq 0 \wedge x \mapsto e' \wedge t = e'\}$$

$$\{x \neq 0 \wedge x \mapsto e' \star \mathrm{list}(e')\}$$
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